Customer Churn Analysis and Retention Strategy Using Data Analytics

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1. Introduction

Customer churn is a major challenge for subscription-based and service-oriented businesses. When customers discontinue using a service, it leads to direct revenue loss and increases customer acquisition costs. Retaining existing customers is often more cost-effective than acquiring new ones.

The objective of this project is to perform an **end-to-end data analysis** on customer churn data to identify key factors influencing churn and to propose data-driven business strategies that can help reduce customer attrition.

This project follows a structured analytics workflow including:

- Data preparation
- Exploratory data analysis
- Statistical analysis
- Insight generation
- Business recommendations

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2. Dataset Description

The dataset used in this analysis is **customer_churn.csv**, which contains customer-level information related to service usage and churn behavior.

Dataset Characteristics:

- **Number of records:** 500+
- **Target variable:** `Churn` (0 = No, 1 = Yes)
- **Data types:** Numerical and categorical

- **Business relevance:** Suitable for churn analysis, customer segmentation, and retention strategy design

Key Columns:

- `Churn` – Indicates whether the customer has churned
- `MonthlyCharges` – Monthly service charges
- `Tenure` – Duration of customer relationship
- Additional service and demographic attributes

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3. Data Cleaning and Preparation

Before analysis, the dataset was cleaned to ensure accuracy and reliability.

Cleaning Steps:

1. **Missing Value Handling**
    - Numerical columns: Missing values were replaced with the column mean.
    - Categorical columns: Missing values were replaced with the most frequent value (mode).
2. **Data Type Validation**
    - Numerical fields were validated for proper numeric formatting.
    - Categorical fields were ensured to be correctly encoded.
3. **Data Integrity Checks**
    - Verified that churn labels contained valid values (0 and 1).
    - Confirmed no empty groups existed for statistical comparison.

This step ensured the dataset was ready for reliable statistical and exploratory analysis.

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4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand patterns, trends, and anomalies in the data.

4.1 Churn Distribution

A count plot was used to visualize the distribution of churned and non-churned customers.

Observation:

- Majority of customers did not churn.
- A smaller but significant proportion of customers churned, indicating a real business problem.

4.2 Monthly Charges Distribution

A histogram with a density curve was used to analyze the distribution of monthly charges.

Observation:

- Monthly charges show a wide range.
- Presence of higher charges suggests potential churn risk among premium customers.

4.3 Tenure vs Churn

A boxplot was used to compare customer tenure between churned and retained customers.

Observation:

- Churned customers generally have shorter tenure.
- Long-term customers are more likely to stay with the service.

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5. Descriptive Statistics

Descriptive statistics were calculated to summarize numerical features.

Metrics Computed:

- Mean
- Median
- Standard Deviation
- Minimum and Maximum values

Group-wise Comparison:

Statistics were calculated separately for:

- Churned customers (`Churn = 1`)
- Non-churned customers (`Churn = 0`)

Key Finding:

- Average monthly charges differ between churned and non-churned customers.
- Tenure is significantly lower for churned customers.

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6. Correlation Analysis

A correlation heatmap was generated using numerical variables to identify relationships among features.

Observations:

- Monthly charges show correlation with other service-related variables.
- Some features move together, indicating interdependencies in customer behavior.
- Correlation analysis helps identify variables useful for churn prediction.

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7. Hypothesis Testing

To statistically validate the observed differences, an **independent t-test** was conducted.

7.1 Hypothesis Formulation

- **Null Hypothesis (H_0):**
  There is no significant difference in monthly charges between churned and non-churned customers.

- **Alternative Hypothesis (H_1):**
  There is a significant difference in monthly charges between churned and non-churned customers.

7.2 Test Methodology

- Test used: **Welch's t-test**
- Reason: Unequal sample sizes (53 churned, 447 non-churned)
- Significance level: $\alpha = 0.05$

7.3 Results

- **Churned customers:** 53
- **Non-churned customers:** 447
- **p-value:** < 0.05

Interpretation:

Since the p-value is less than the significance level, the null hypothesis is rejected.

Conclusion:
There is a statistically significant difference in monthly charges between churned and non-churned customers.

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8. Confidence Interval Estimation

A **95% confidence interval** was calculated for the mean monthly charges.

Purpose:

- To estimate the range within which the true average monthly charge lies.
- To quantify uncertainty in the estimate.

Result:

The confidence interval provides a reliable range for average charges and supports pricing-related insights.

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9. Key Insights

From the analysis, the following insights were derived:

1. Customers with higher monthly charges are more likely to churn.
2. Short-tenure customers show higher churn rates.
3. Long-term customer retention is associated with reduced churn.
4. Pricing and tenure are critical churn drivers.
5. Statistical testing confirms pricing differences are significant.

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10. Business Recommendations

Based on analytical findings, the following recommendations are proposed:

1. **Target High-Charge Customers**
    - Offer discounts or loyalty plans to customers with high monthly charges.
2. **Improve Early Customer Engagement**
    - Enhance onboarding for new customers to reduce early churn.

3. **Promote Long-Term Contracts**
    - Incentivize customers to move from short-term to long-term plans.
4. **Churn Monitoring System**
    - Track high-risk customers using tenure and pricing indicators.

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11. Conclusion

This project successfully applied statistical and exploratory data analysis techniques to identify key drivers of customer churn. The analysis demonstrated that pricing and tenure significantly influence customer retention. The insights and recommendations derived from this study can help businesses reduce churn, improve customer satisfaction, and enhance long-term profitability.